

# MAXIMILIAN VINCENTIUS

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## EDUCATION

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| <b>Sekolah Ciputra, IB Diploma</b> <ul style="list-style-type: none"><li>Higher Level (HL): Computer Science, Mathematics AA</li></ul> <b>UCLA</b> <ul style="list-style-type: none"><li>Intensive program focusing on Game Design and interactive systems, earning 4.0 UCLA.</li></ul> | <b>Expected 2027</b><br><br><b>July 2026</b> |
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## TECHNICAL EXPERTISE

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| <b>Programming Languages:</b> Python, C# (Unity), C++, JavaScript, HTML, CSS                                                            |
| <b>Frameworks &amp; Tools:</b> Unity, Git, Autodesk Maya, Adobe Substance Painter, Node.js, NoSQL, SQL, Supabase, Tailwind              |
| <b>Key Competencies:</b> Full-Stack Web Development, 3D Modeling & Texturing, Game Physics Simulation, Interactive Systems Architecture |

## ENGINEERING & SYSTEMS

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| <b>Founder – pointxtech.com (Independent Software Studio)</b> <ul style="list-style-type: none"><li>Led development and deployment of multi-platform systems (web, game, simulation) through a unified pipeline, achieving seamless integration and efficient delivery.</li><li>Phyxhub.com (Physics Education Initiative, Aug 2025 – Present): Built a full-stack IB Physics Learning Hub (Vite, React, SQL) featuring a mastery-gated content-locking system and a Figma-designed UI, alongside a library of simplified physics guides and visual aids.</li><li>Fractals of Entropy: Engineered a 3D action-strategy game engine with procedural generation, AI pathfinding, and based combat systems, shipping builds across Windows, Android (Google Play), and mobile.</li><li>Metro Circuit: Built a Unity-based simulation with NPC navigation, traffic systems, and real-time world interaction, shipping builds across Windows, Android (Google Play), and mobile</li></ul> | <b>June 2025 – Present</b>            |
| <b>Lead System Architect – Swastisvarna.org (Non-profit Infrastructure)</b> <ul style="list-style-type: none"><li>Rebuilt legacy web infrastructure into a Node.js + Supabase full-stack system deployed on a VPS, including a custom CMS that empowered non-technical staff and cut manual administrative workload by approximately 5 hours per week.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>January 2024 – Present</b>         |
| <b>Founder – 3D Game Creator Club</b> <ul style="list-style-type: none"><li>Mentored 16 students in Unity development, 3D modeling, and game systems design, fostering practical skills and collaborative project completion.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>September 2025 – November 2025</b> |

## RESEARCH & COMPUTATIONAL INVESTIGATIONS

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| <b>Complex Fourier Analysis of Instrument Timbre (Mathematics AA HL)</b> <ul style="list-style-type: none"><li>Decomposed recorded piano and flute/saxophone tones into complex Fourier coefficients(<math>C_n</math>) to test how well the coefficients distinguish instruments at equal pitch across dynamics. Applied the Nyquist sampling theorem, reconstructed recordings from the extracted coefficients, and scored accuracy via Mean Squared Error against the originals, demonstrating applied signal-processing and complex-analysis skills.</li></ul>                                   | <b>July 2026 – August 2026</b> |
| <b>Eddy-Current Damping of a Pendulum (Physics HL)</b> <ul style="list-style-type: none"><li>Built a Waltenhofen-style pendulum apparatus to test how magnetic field strength (B) affects the damping coefficient (<math>\gamma</math>) of an eddy-current brake. Measured B with a Gaussmeter across five settings, extracted <math>\gamma</math> from <math>\ln(A)</math>-vs-swing-number graphs via video analysis, and confirmed the predicted <math>\gamma \propto B^2</math> relationship from Faraday's and Lenz's laws, providing empirical insight into electromagnetic braking.</li></ul> | <b>June 2026 – July 2026</b>   |
| <b>Golden-Ratio Wave Cascades in Real-Time Ocean</b> <ul style="list-style-type: none"><li>Tested whether golden-ratio frequency spacing across Gerstner-wave cascades reduces large-scale periodicity in a real-time Unity HLSL compute-shader ocean, versus a single-cascade baseline and integer-ratio cascades of equal wave count. Implemented the cascades as a GPU compute shader and quantified visual realism via the Structural Similarity Index (SSIM), contributing to optimized, artifact-free real-time water rendering</li></ul>                                                     | <b>May 2026 – August 2026</b>  |

## AWARDS & HONORS

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| <b>Technology:</b> Gold Medal (6th), Intl. STEM Coding Olympiad (2023); Gold Medal, NCSS Grok Python (2022, 2023). |
| <b>Mathematics:</b> Silver Medal, Intl. Online Math Challenge (2024); Gold Medal, ASMO Mathematics (2022).         |
| <b>Science:</b> Silver Medal, ASMO Science (2022); Honorable Mention, Intl. STEM Math Olympiad (2023).             |
| <b>Writing/Leadership:</b> Double Gold Medalist (Writing), World Scholar's Cup; TOC Qualifier (2024).              |